CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

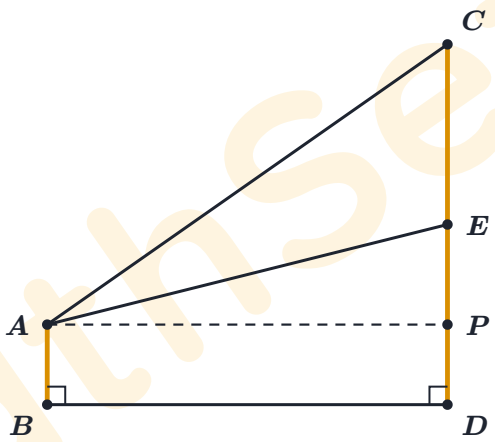
Chapter 22: Heights and Distances

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Problems with two triangles

1. A vertical tower AB stands 100 m from the foot D of a building CD. E is the mid-point of CD. The angle of depression of the top A of the tower is 35° from C and 14° from E. Using tangent values rounded to two decimal places, find the height of (a) the tower (b) the building. ICSE 2025, 5 marks



SOLUTION

Draw AP horizontal, meeting CD at P. Then $AP = BD = 100$ m and $PD = AB$. The angles of depression equal $\angle CAP = 35^\circ$ and $\angle EAP = 14^\circ$ (alternate angles).

From the tables, $\tan 35^\circ = 0.70$ and $\tan 14^\circ = 0.25$.

$CP = 100 \tan 35^\circ = 70$ m and $EP = 100 \tan 14^\circ = 25$ m, so $CE = 70 - 25 = 45$ m.

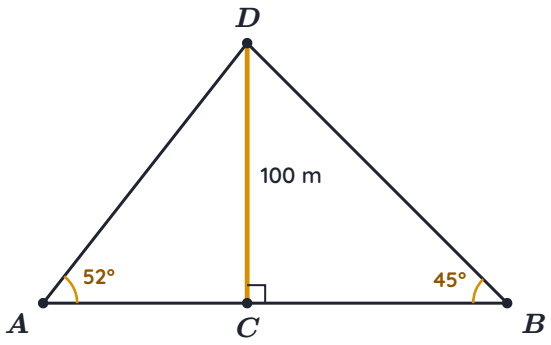
E is the mid-point of CD, so $CD = 2 \times 45 = 90$ m and $ED = 45$ m.

$AB = PD = ED - EP = 45 - 25 = 20$ m

Answer: (a) tower 20 m (b) building 90 m

2. A tree CD is 100 m high. From two points A and B on the ground on opposite sides of the tree, its top D is seen at angles of elevation of $52°$ and $45°$. Find the distance AB to the nearest metre.

ICSE 2024, 5 marks



SOLUTION

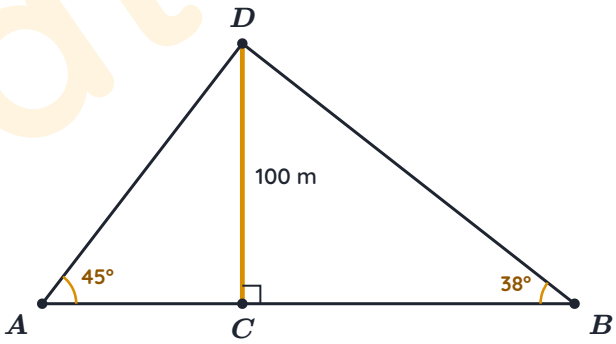
In $\triangle ACD$: $\tan 52° = \dfrac{100}{AC}$. From the tables $\tan 52° = 1.2799$, so $AC = \dfrac{100}{1.2799} \approx 78.13$ m.

In $\triangle BCD$: $\tan 45° = \dfrac{100}{CB}$, so $CB = 100$ m.

$AB = AC + CB = 78.13 + 100 = 178.13$

Answer: $AB \approx 178$ **m**

3. From the top D of a tower CD 100 m high, two ships A and B on opposite sides of the tower are seen at angles of depression of $45°$ and $38°$. The ships and the foot of the tower are in one straight line. Find the distance between the ships to the nearest metre. ICSE 2023, 4 marks



SOLUTION

The angles of elevation of D from the ships equal the angles of depression (alternate angles).

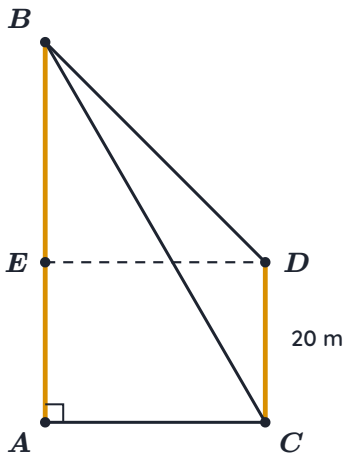
$AC = \dfrac{100}{\tan 45°} = 100$ m. From the tables $\tan 38° = 0.7813$, so $CB = \dfrac{100}{0.7813} \approx 127.99$ m.

$AB = 100 + 127.99 = 227.99$

Answer: about 228 m

4. From the top of a cliff, the top and the bottom of a tower 20 m high are seen at angles of depression of 45° and 60°. Find (a) the height of the cliff (b) the distance between the cliff and the tower.

ICSE 2020, 4 marks



SOLUTION

Let $AB = h$ be the cliff, $CD = 20$ m the tower and $AC = d$. Draw DE horizontal to meet the cliff at E.

From the bottom of the tower: $\tan 60^\circ = \dfrac{h}{d}$, so $h = \sqrt{3}\,d$.

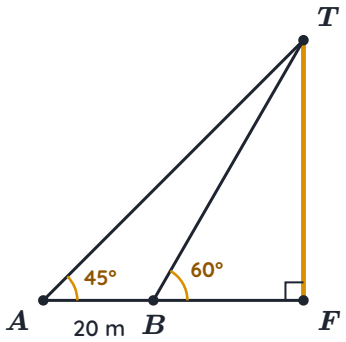
From the top of the tower: $\tan 45^\circ = \dfrac{h - 20}{d}$, so $h - 20 = d$.

$\sqrt{3}\,d - 20 = d$, so $d = \dfrac{20}{\sqrt{3} - 1} = 10(\sqrt{3} + 1) \approx 27.32$ m and $h = d + 20 \approx 47.32$ m.

Answer: (a) about 47.32 m (b) about 27.32 m

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5. A man sees the top of a tower at an angle of elevation of $45°$. He walks 20 m straight towards the tower, and the angle of elevation becomes $60°$. Find the height of the tower correct to 2 significant figures. ICSE 2019, 4 marks



SOLUTION

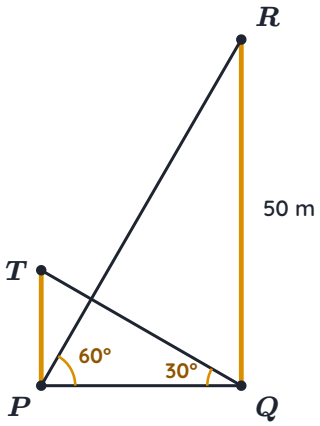
Let the height be h and let the man start x m from the tower.

$\tan 45° = \dfrac{h}{x}$, so $x = h$. $\tan 60° = \dfrac{h}{x - 20}$, so $h = \sqrt{3}(h - 20)$.

$h(\sqrt{3} - 1) = 20\sqrt{3}$, so $h = \dfrac{20\sqrt{3}}{\sqrt{3} - 1} = 10\sqrt{3}(\sqrt{3} + 1) = 30 + 10\sqrt{3} \approx 47.32$

Answer: about 47 m

6. QR is a tower 50 m high and PT is another tower, with P and Q on level ground. From P the angle of elevation of R is $60°$, and from Q the angle of elevation of T is $30°$. Find the height of PT to the nearest metre. ICSE 2018, 4 marks



SOLUTION

In $\triangle PQR$: $\tan 60° = \dfrac{50}{PQ}$, so $PQ = \dfrac{50}{\sqrt{3}}$ m.

In $\triangle TPQ$: $\tan 30° = \dfrac{PT}{PQ}$, so $PT = \dfrac{50}{\sqrt{3}} \times \dfrac{1}{\sqrt{3}} = \dfrac{50}{3} \approx 16.67$

Answer: about 17 m

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7. From the top of a lighthouse 60 m high, two ships on opposite sides of it are seen at angles of depression of 60° and 45°. Find the distance between the ships to the nearest whole number.

ICSE 2017, 4 marks

SOLUTION

The angle of elevation of the top from each ship equals the angle of depression.

Distance of the first ship $= \dfrac{60}{\tan 60^\circ} = \dfrac{60}{\sqrt{3}} = 20\sqrt{3} \approx 34.64$ m

Distance of the second ship $= \dfrac{60}{\tan 45^\circ} = 60$ m

Total $= 34.64 + 60 = 94.64$

Answer: about 95 m

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